



Graphing direct proportion

Task A: Features of graphs

1) Which of the following tables show that A and B are in direct proportion? Give a reason for your answers.

a)

A	B
3	96
12	384

$\times 4$ $\times 4$ ✓

c)

A	B
152	60.8
6	1.8

$\times 0.4$ $\times 0.3$ ✗

b)

A	B
16	12
4	2.9

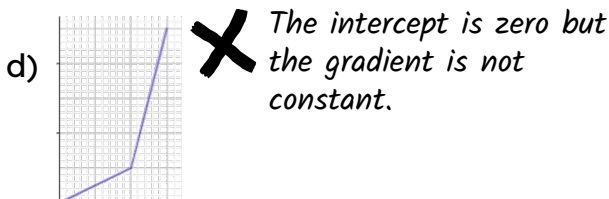
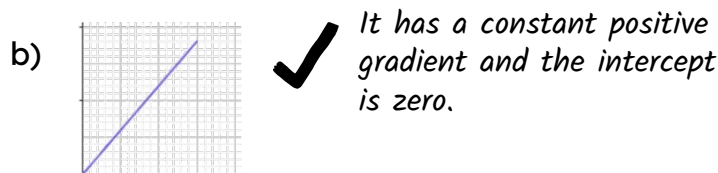
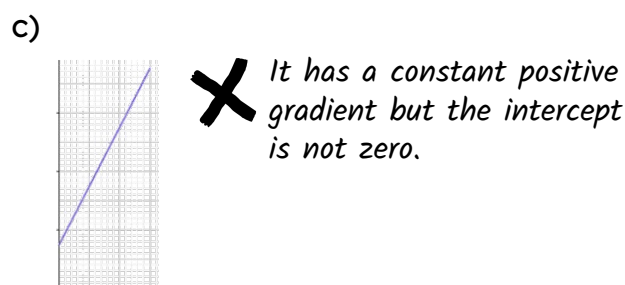
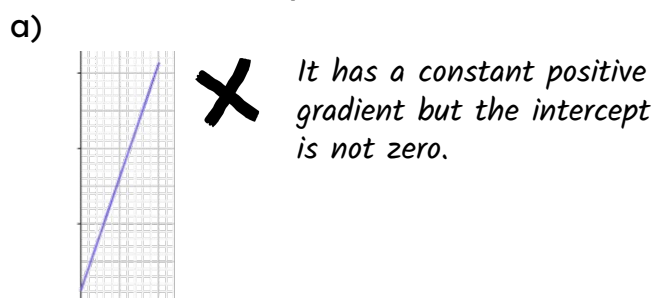
$\times 0.25$ or $\div 4$ $\times \frac{29}{120}$ ✗

d)

A	B
17	30.6
45	81

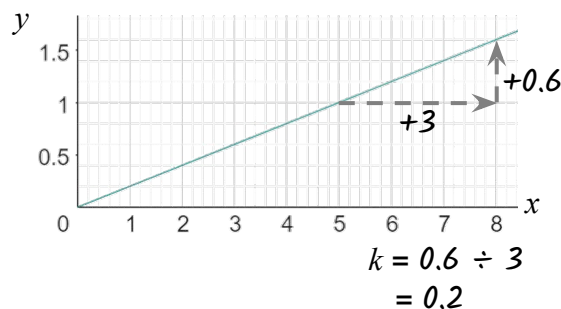
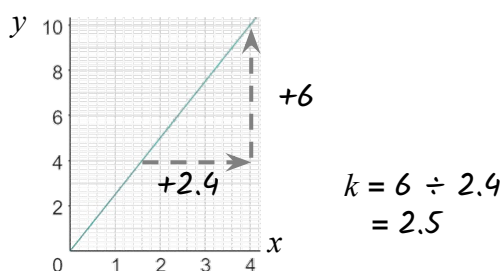
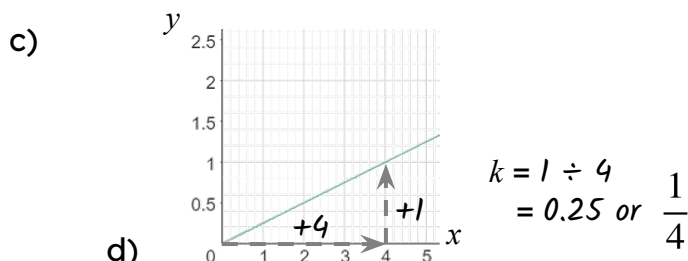
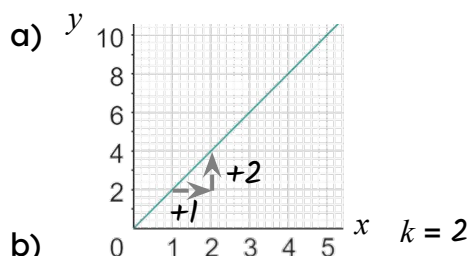
$\times 1.8$ $\times 1.8$ ✓

2) Which of the following graphs show two variables that are in direct proportion? Give a reason for your answers.



Task B: Graphs representing direct proportion

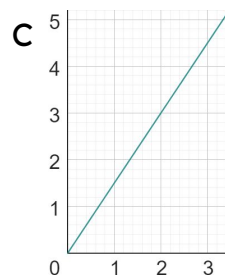
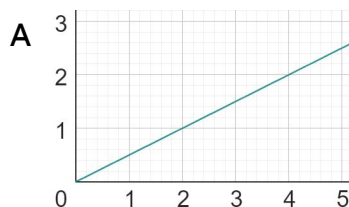
1) Calculate the value of k (constant of proportionality) from the following graphs:



2) Match each table, equation and graph.

a)

x	0	2	4	6	8
y	0	3	6	9	12



i) $y = 3x$

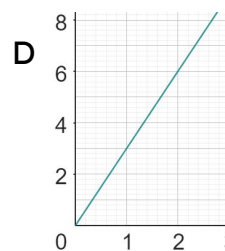
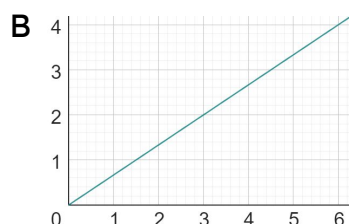
b)

x	0	1	4	7	10
y	0	3	12	21	30

ii) $y = 0.5x$

c)

x	0	6	18	24	30
y	0	4	12	16	20



iii) $y = 1.5x$

iv) $y = \frac{2}{3}x$

d)

x	0	2	8	12	18
y	0	1	4	6	9

- a) C iii)
b) D i)
c) B iv)
d) A ii)